

# Soumadip Biswas, *Ph.D.*


**Summary**

Software Engineer and Ph.D. with a foundation in distributed systems, networking, and algorithms. Experienced in high-performance C++ at Citrix and currently building modular automation tools and AI-assisted workflows. A builder and lifelong learner focused on bridging algorithmic research with practical, scalable software solutions to solve complex SDE challenges.

**Contact Information**

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**Education**

**Indian Institute of Technology Kharagpur**, Ph.D. in Computer Science (2014–2023)
Thesis: Algorithms for Handover Management in Dense LTE Networks under Vehicular Mobility
**Indian Institute of Technology Kharagpur**, M.Tech. in Information Technology (2010–2012)
Thesis: Geospatial Services in Cloud Infrastructure
**West Bengal University of Technology**, B.Tech. in Computer Science & Engineering (2006–2010)

**Technical Focus**

Distributed Systems & Algorithms · Cellular Networking · Operating Systems · Graph Theory

**Experience**

- **Associate Professor, Institute of Engineering and Management, Kolkata (2022—Present)**
  - **Engineering Automation:** Architected and deployed a suite of internal tools in Python and Go to automate academic workflows for 250+ students and 10+ faculty members.
  - **AI-Driven Evaluation:** Developed an automated term project reviewer using the Gemini API to evaluate student Git repositories, streamlining feedback loops and ensuring code quality.
  - **Video Processing Research:** Engineered a framework for Indian Sign Language (ISL) recognition using OpenCV2 and Hugging Face models; designed custom data structures to graphically represent finger movements.
  - **Content Search Optimization:** Developed algorithms for efficient metadata indexing to enable high-speed content searching within large-scale video datasets.
  - **Modular Design:** Implemented all internal tooling with a focus on extensibility and modularity, ensuring long-term maintainability and ease of containerization.
- **PhD Researcher (Distributed Systems & Networking), IIT Kharagpur (2014—2023)**
  - **Distributed Algorithm Design:** Engineered high-performance mobility management algorithms for dense LTE/vehicular networks, effectively reducing spurious handovers and latency.
  - **Large-Scale Simulation:** Developed and implemented scalable system prototypes in **C++** using the NS3/LENA framework, processing massive vehicular mobility traces to validate system performance.

  - **Performance Optimization:** Optimized network resource allocation and load-balancing protocols, achieving significant improvements in user-association efficiency.
  - **Peer-Reviewed Excellence:** Published high-impact research in top-tier venues including **IEEE INFOCOM** and **Elsevier Computer Networks**.
- **Software Engineer II, (Promoted from SE I) Citrix R&D India Pvt. Ltd. (2012—2014)**
  - Developed and optimized XenApp features using C++ and C#, improving system stability across multi-node deployments.
  - Reduced remote session latency and memory overhead through targeted performance tuning.

## Technical Projects

**Home Lab Infrastructure & Automation**

- **Edge Computing:** Configured a Raspberry Pi-based home lab accessible via reverse proxy to bypass CGNAT, enabling secure remote management.
- **Containerization:** Deployed and orchestrated a suite of services using Docker, including custom-built tools for PDF manipulation, image processing, and document organization.
- **Micro-tooling:** Developing lightweight, container-based Go and Python utilities for daily task automation, focusing on concurrency and low resource footprints.
- **Automated Academic Suite**
  - Built a cross-platform attendance reporting and automated exam evaluation engine, reducing manual grading overhead by approximately 40%.

## Achievements

**GATE 2010:** All India Rank 330 (Top 0.31% of candidates)
**MHRD Scholarships:** Awarded for Ph.D. (2014–19) and M.Tech. (2010–12)

## Technical Skills

- **Languages:** C++, Python, C#, Go (Familiar), Shell Scripting, Java, SQL.
- **AI & Media:** OpenCV2, Hugging Face Transformers, Gemini API, Video Indexing
- **Infrastructure & Tools:** Docker, Linux/Unix, Git, NS3, MPI, Reverse Proxies
- **Core Expertise:** Distributed Systems, Scalable Algorithms, Network Optimization

## Selected Publications

- S. Biswas, S. Chakraborty, and A. Gupta, "Reducing Spurious Handovers in Dense LTE Networks based on Signal Strength Look-ahead,"
  textitIEEE WiMob, 2018.
- S. Biswas, A. Gupta, and S. Chakraborty, "A Framework for eNB Load Balancing in Dense LTE Networks," *IEEE PIMRC*, 2019.
- S. Biswas, S. Chakraborty, and A. Gupta, "Poster: Balanced User Association in Dense LTE Networks," *IEEE INFOCOM*, 2019.
- S. Biswas, A. Gupta, and S. Chakraborty, "Load-balanced User Associations in Dense LTE Networks," *Computer Networks*, Vol. 189, 2021.

## References


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